BEING AFFINE IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's own aside 'In fact A = Gamma(X, O_X) of course.' and the phrase 'and we win' are quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Base-change stability and Zariski locality of affineness. Morphisms, Lemmas morphisms-lemma-base-change-affine and morphisms-lemma-characterize-affine, invoked at proof lines 6149-6152. Supplies the two preliminary halves of fpqc locality.
- Quasi-compactness and separatedness descend along a flat surjective map of affines. Same chapter, Lemmas lemma-descending-property-quasi-compact and lemma-descending-property-separated, invoked at proof lines 6161-6163. Makes f_* O_X quasi-coherent, so that it is the tilde of an R-algebra A.
- Flat base change for quasi-coherent cohomology, and the canonical morphism into an affine scheme. Cohomology of Schemes, Lemma coherent-lemma-flat-base-change-cohomology and Schemes, Lemma schemes-lemma-morphism-into-affine, invoked at proof lines 6168-6177. Gives A' = R' tensor_R A and produces the comparison morphism X -> Spec(A).
- Isomorphisms descend, and the general descending-properties principle. Same chapter, Lemmas lemma-descending-property-isomorphism and lemma-descending-properties-morphisms, invoked at proof lines 6179-6181. Concludes that X -> Spec(A) is an isomorphism and finishes the locality statement.

Source reference. Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KP: sources/stacks-descent/pinned-original-source.tex, source lines 5624–5624. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02L4: sources/stacks-descent/pinned-original-source.tex, source lines 6132–6132. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KQ: sources/stacks-descent/pinned-original-source.tex, source lines 5688–5688. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KU: sources/stacks-descent/pinned-original-source.tex, source lines 5798–5798. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01SD: sources/stacks-chow/tags.txt, source lines 2334–2334. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01S8: sources/stacks-chow/tags.txt, source lines 2329–2329. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01I1: sources/stacks-chow/tags.txt, source lines 1962–1962. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01LC: sources/stacks-chow/tags.txt, source lines 2081–2081. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is affine" is fpqc local on the base.*

Proof. A base change of an affine morphism is affine, see Morphisms, Lemma 01SD. Being affine is Zariski local on the base, see Morphisms, Lemma 01S8. Finally, let $g : S' \to S$ be a flat surjective morphism of affine schemes, and let $f : X \to S$ be a morphism. Assume that the base change $f' : X' \to S'$ is affine. In other words, X' is affine, say $X' = \mathrm{Spec}(A')$. Also write $S = \mathrm{Spec}(R)$ and $S' = \mathrm{Spec}(R')$. We have to show that X is affine.

By Lemmas 02KQ and 02KU we see that $X \to S$ is separated and quasi-compact. Thus $f_*\mathcal{O}_X$ is a quasi-coherent sheaf of $\mathcal{O}_S$-algebras, see Schemes, Lemma 01LC. Hence $f_*\mathcal{O}_X = \widetilde{A}$ for some R-algebra A. In fact $A = \Gamma(X, \mathcal{O}_X)$ of course. Also, by flat base change (see for example Cohomology of Schemes, Lemma 02KH) we have $g^*f_*\mathcal{O}_X = f'_*\mathcal{O}_{X'}$. In other words, we have $A' = R' \otimes_R A$. Consider the canonical morphism

$$X \longrightarrow \mathrm{Spec}(A)$$

over S from Schemes, Lemma 01I1. By the above the base change of this morphism to S' is an isomorphism. Hence it is an isomorphism by Lemma 02L4. Therefore Lemma 02KP applies and we win. □